PROJECT 23

Fabric-Picking Mechanism Research

MAS Kreeda Vaanavil (MAS Active Group)
2019 (3-month Internship) · Autonomation / Automation Engineering Intern

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PROJECT SUMMARY

Researched and developed concept designs for automatically picking individual fabric pieces from cut-bundle stacks — the key unsolved challenge preventing full automation of the garment bonding process. Two methods were investigated: Bernoulli air-jet separation and fabric-stretch separation. Both were taken to concept-design level with SolidWorks drawings.

OBJECTIVES

- Find a reliable method to separate and pick single fabric layers from a cut-bundle stack.
- Research and evaluate Bernoulli air-jet separation.
- Research and evaluate fabric-stretch-based separation.
- Produce concept CAD drawings and design recommendations.

HOW IT WORKS

Method 1 (Bernoulli): An air jet creates a low-pressure zone above the top fabric layer, lifting it slightly so a gripper or suction cup can pick it. Method 2 (Fabric Stretch): A clamp holds the bundle; a lifting head grips and stretches the top layer away from the bundle, separating it by exploiting the fabric's elasticity. Both methods were designed, modelled in SolidWorks, and compared for feasibility.

KEY COMPONENTS

Component	Specification / Function
Air Nozzle + Compressed Air	Bernoulli method — lifts top fabric layer
Suction Cup / Gripper	Bernoulli method — picks separated layer
Fabric Clamp	Stretch method — holds the cut-bundle stack
Lifting Head	Stretch method — grips and stretches top layer
Pneumatic Actuators	Both methods — movement and control

TECHNICAL SPECIFICATIONS

Parameter	Value	Notes
Method 1	Bernoulli air-jet separation	Contactless lift
Method 2	Fabric-stretch separation	Uses fabric elasticity
Design stage	Concept design + SolidWorks drawings	—

MY CONTRIBUTION

- Identified fabric-picking as the key automation gap in the garment process.
- Researched and analysed both separation methods.
- Produced concept CAD designs and SolidWorks drawings for both approaches.
- Wrote design recommendations for the next-generation machine.

CHALLENGES & SOLUTIONS

Fabric layers in a cut bundle cling together due to static, moisture, and fibre interlocking. Neither method was fully proven in the internship period — both remain viable directions for further development.

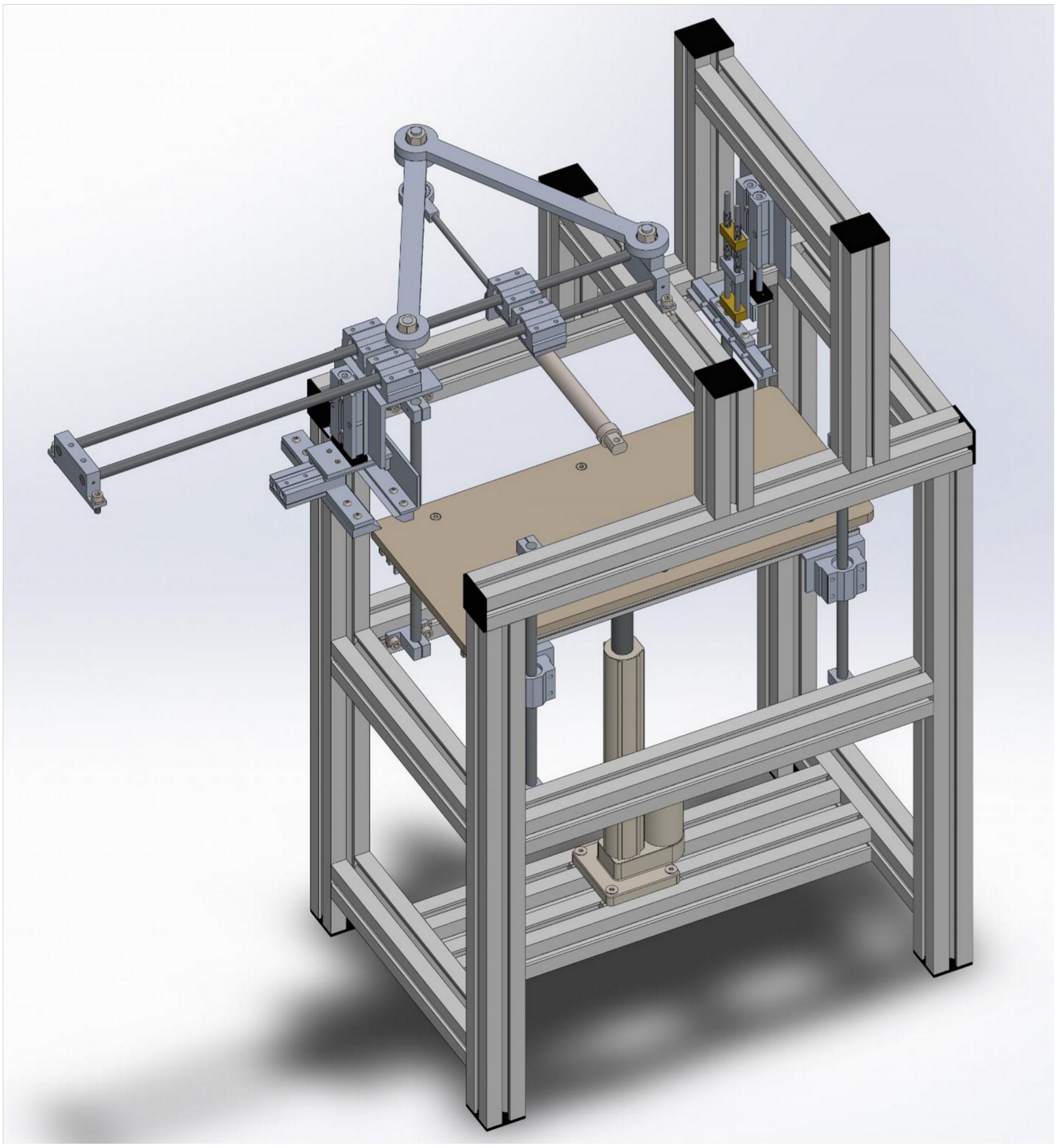
RESULTS

Research report and SolidWorks concept drawings for both separation methods produced and submitted. Provided a clear engineering roadmap for full automation of the garment bonding process.

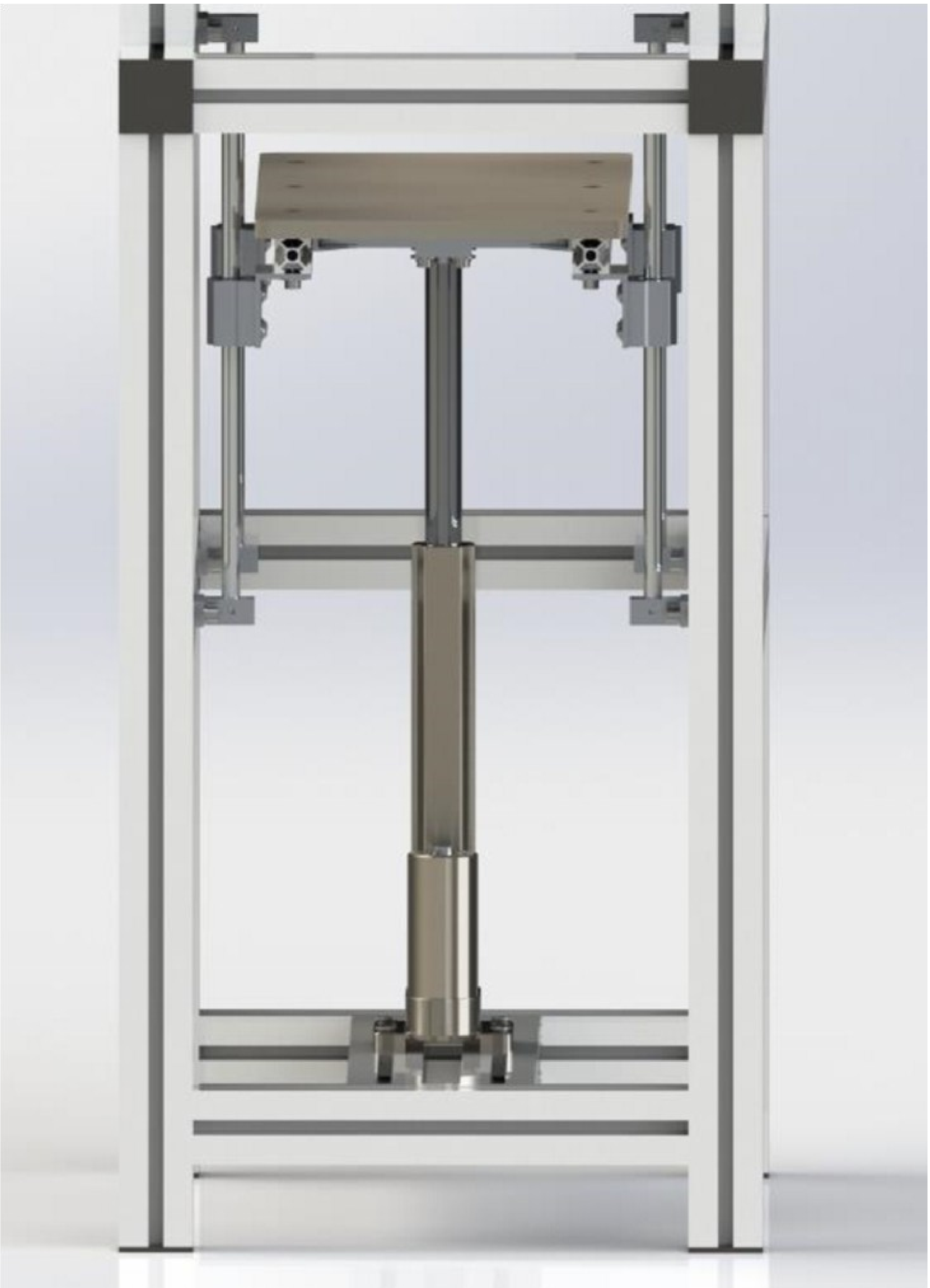
TOOLS & SOFTWARE USED

Tool / Software	Used For
SolidWorks	Concept CAD design for both methods
Pneumatic Design	Circuit concepts for actuation

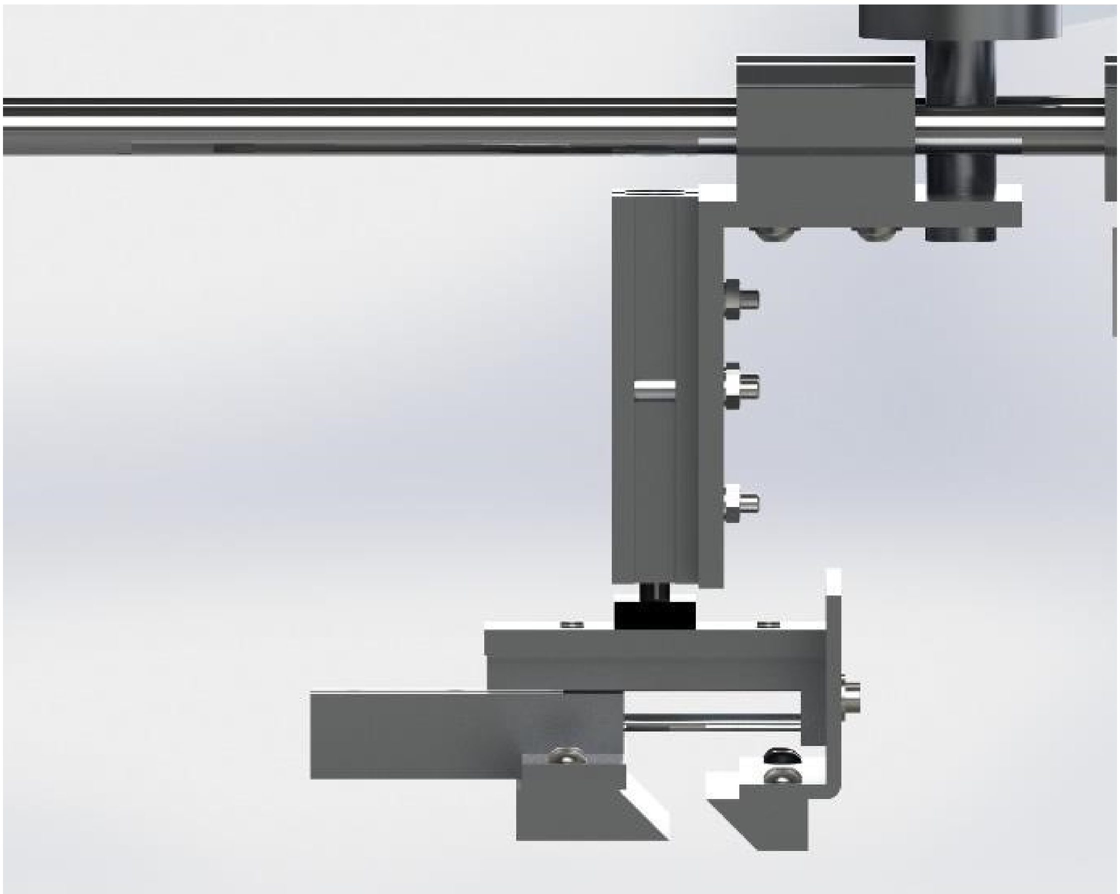
IMAGES & TECHNICAL DRAWINGS



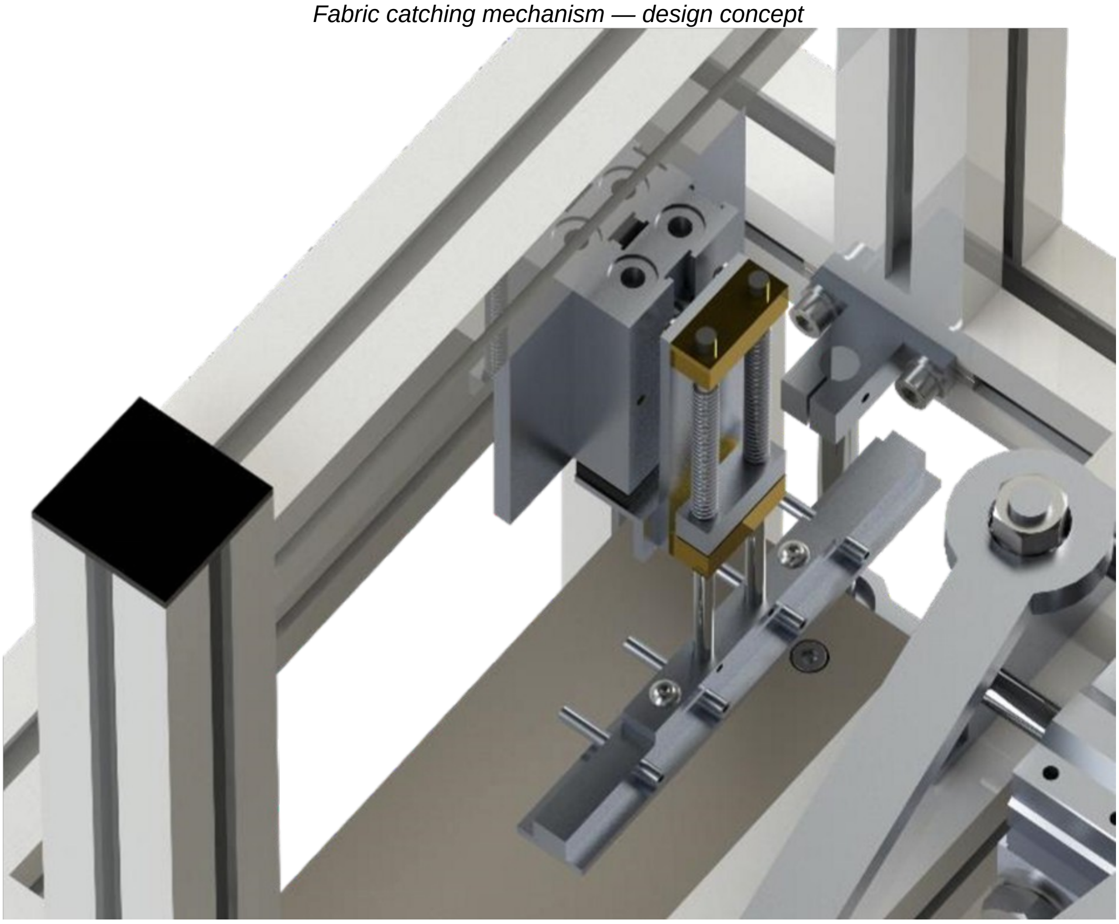
Method 2: Fabric stretching property — concept overview



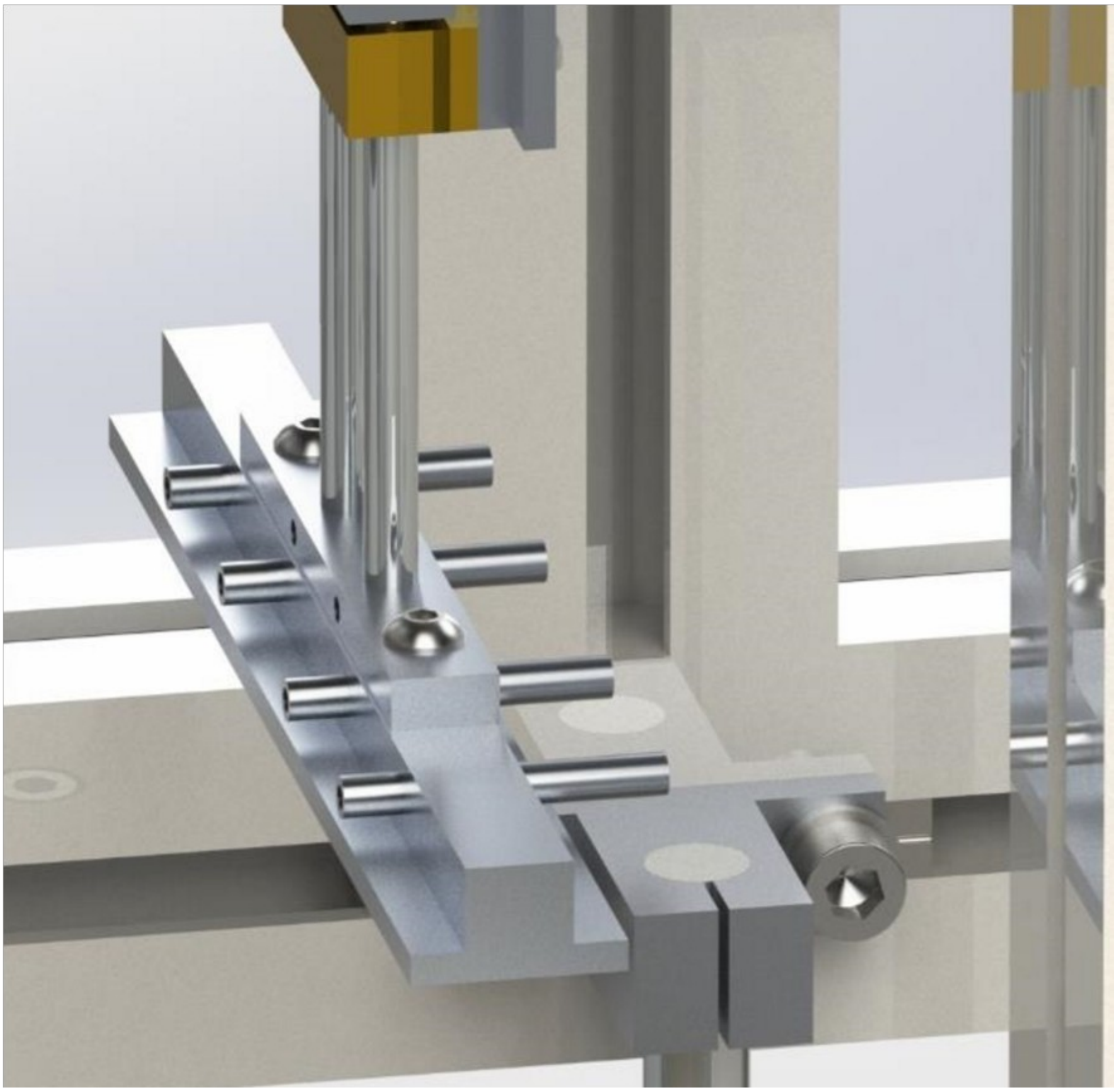
Cut-cake lifting part — design concept



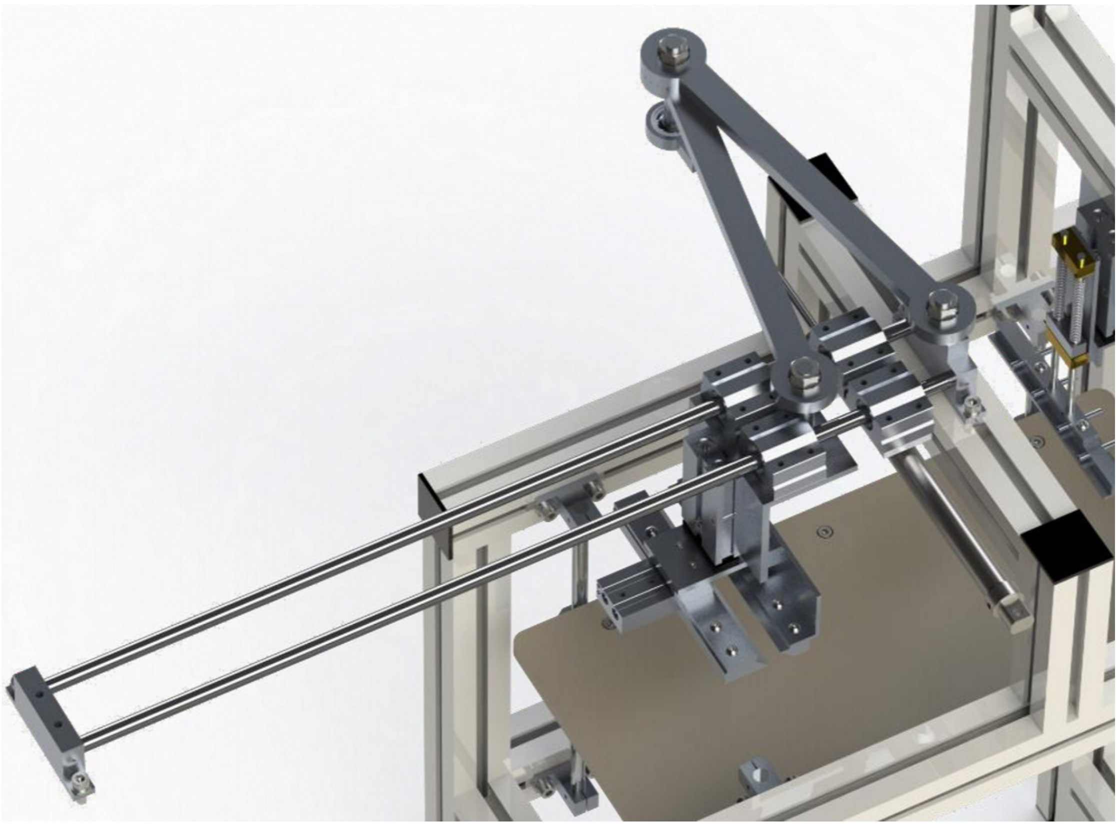
Fabric catching mechanism — design concept



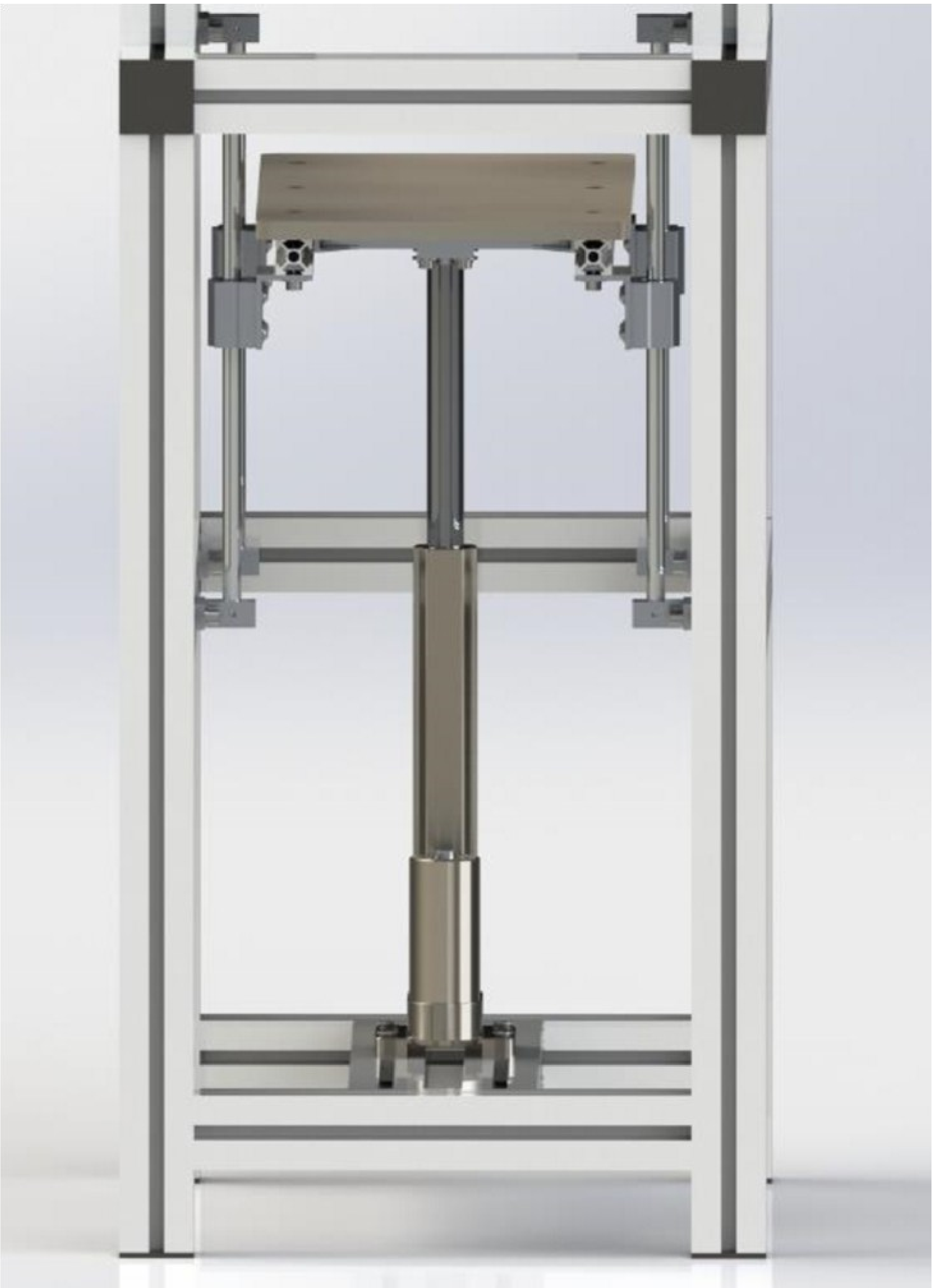
Fabric clamping part — design concept



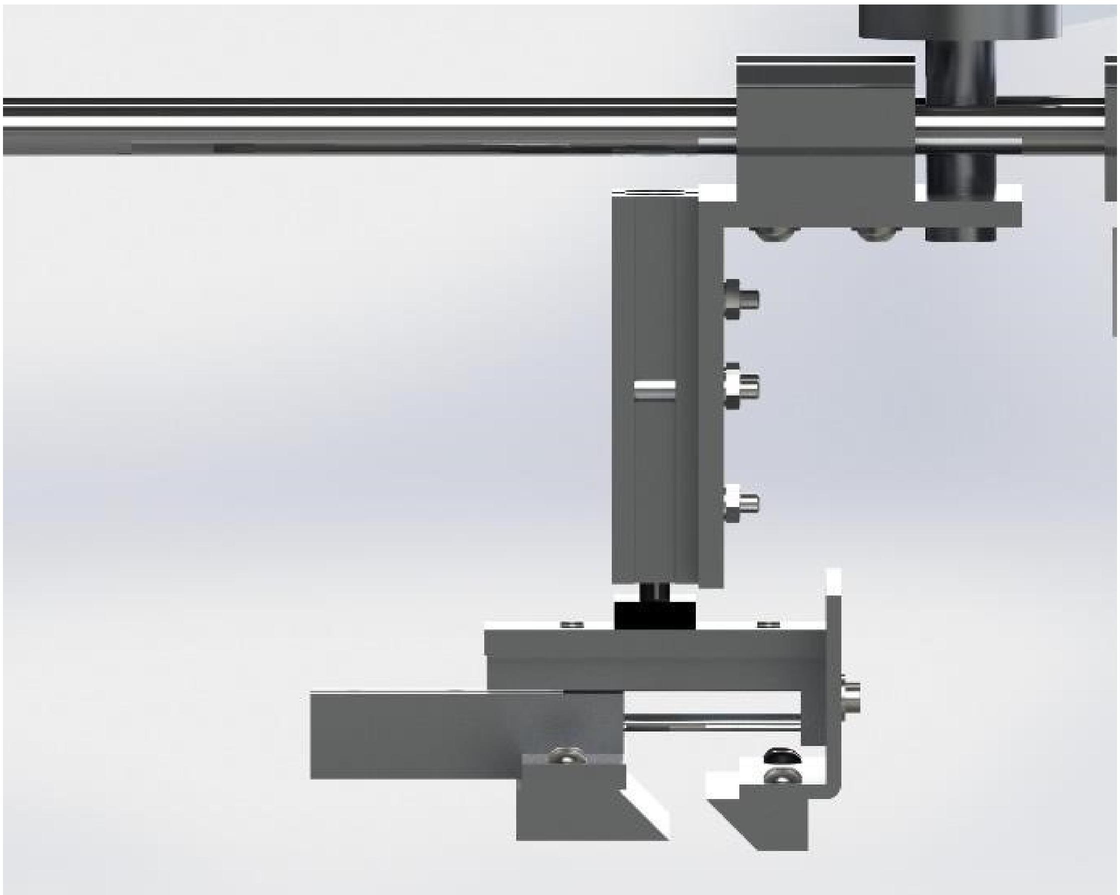
Surface cleaning part — design concept



Fabric moving part — design concept



Cut-cake lifting — alternative design



Fabric catching mechanism — alternative design